Nilai UNIVERSITY

BACHELOR IN INFORMATION TECHNOLOGY (HONOURS)

FINAL EXAMINATION
FEBRUARY 2025

Course: EC3283 (Cybersecurity)
Lecturer: Muhammad Hamizan Johari

Time: 9.00 am – 12.00 pm (3 hours)
Date: 23 May 2025

Instructions:

Answer **ALL** questions.

The maximum number of marks is 100.

This question paper consists of **3** printed pages.
(excluding front cover)

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1. a. Briefly explain the challenges met by computer security. **(6 MARKS)**

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b. Briefly discuss the following terms regarding computer security.

i. Vulnerability **(2 MARKS)**

ii. Threat **(2 MARKS)**

iii. Attack **(2 MARKS)**

c. Identify the access control policy based on the scenario given below.

i. An employee shares a confidential project folder with a coworker by manually setting folder permissions. **(2 MARKS)**

ii. A classified government database only allows access based on security clearance levels set by the system, not by users. **(2 MARKS)**

d. A large financial corporation maintains and manages sensitive information, including employee salary details, confidential client financial portfolios, and internal audit reports. **(9 MARKS)**
This corporation handles critical financial data that must be properly secured to ensure trust, compliance, and operational success.

Within the organization, there are three high-ranking roles (an HR Manager, a Finance Manager, and an Internal Auditor).
Each role is responsible for specific sensitive documents: an employee salary file, a confidential client file, and an internal audit file.

The HR Manager, Finance Manager, and Internal Auditor must be assigned to the appropriate files with the correct read, write, and execute permissions.

2. a. Briefly describe the example of cases based on the following terms.

i. Identity theft **(2 MARKS)**

ii. Cyber stalking **(2 MARKS)**

iii. Investment fraud **(2 MARKS)**

b. Describe the ways to protect our identity from identity theft. **(6 MARKS)**

c. Encrypt the plaintext, "PADHMASHREE" using the Caeser Cipher with shift key of 4. **(4 MARKS)**

d. **(9 MARKS)**

Table 1: Number assignment to each alphabet.

A = 0	B = 1	C = 2	D = 3	E = 4	F = 5	G = 6
H = 7	I = 8	J = 9	K = 10	L = 11	M = 12	N = 13
O = 14	P = 15	Q = 16	R = 17	S = 18	T = 19	U = 20
V = 21	W = 22	X = 23	Y = 24	Z = 25	-	-

Based from the table above, use Vinegere Cipher to encrypt the following information.

Plaintext: "LOGHORIZON"
Key: "CODE"

3. a. Briefly discuss the following key aspects of privacy in cybersecurity.

   i. Data minimization **(2 MARKS)**

   ii. Transparency **(2 MARKS)**

   iii. User control **(2 MARKS)**

b. Briefly explain the following key aspects of individual's social implication in cybersecurity.

   i. Legal & Ethical Considerations **(2 MARKS)**

   ii. Social Behavior & Cyber Etiquette **(2 MARKS)**

   iii. National Security & Cyber Warfare **(2 MARKS)**

c. An online illegal trader has been caught by the authorities, and the legal team requires evidence to be extracted from his mobile phone to be brought into the judiciaries court. As the police IT investigator, apply **THREE (3)** mobile forensic techniques to obtain the evidence. **(9 MARKS)**

d. Identify the legal aspects of cybersecurity based on the given laws and regulatory acts. **(4 MARKS)**

   i. Nepal's Electronic Transaction Act 2063 addresses various aspects of electronic transactions, including data protection and privacy.

   ii. ISO/IEC 27001 by International Organization for Standardization (ISO) and International Electrotechnical Commission (IEC)

4. a. Briefly explain the following terms regarding ethical issues in cybersecurity.
   i. Responsible disclosure of vulnerabilities **(2 MARKS)**
   ii. Data protection and integrity **(2 MARKS)**
   iii. Dual-use technologies **(2 MARKS)**

   b. Briefly discuss **THREE (3)** ethical issues that can affect corporations. **(6 MARKS)**

   c. Identify the types of malwares based on the definition given below.
   i. Software that collects information from a computer and transmits it to another system by monitoring keystrokes, screen data and/or network traffic; or by scanning files on the system for sensitive information. **(2 MARKS)**
   ii. A type of virus that uses macro or scripting code, typically embedded in a document, and triggered when the document is viewed or edited, to run and replicate itself into other such documents. **(2 MARKS)**

Social engineering is a manipulation technique that exploits human error or trust to gain private information, access, or valuables. Instead of directly attacking systems through technical means, attackers use psychological tricks to deceive individuals into breaking security protocols or revealing sensitive data. Common forms of social engineering include phishing emails, pretexting (creating a fake scenario), baiting (offering something enticing), and tailgating (following someone into a restricted area). Because it targets human behavior rather than system vulnerabilities, social engineering can be one of the most effective and dangerous methods of attack.

   d. Based from the paragraph above, suggest **THREE (3)** ways to protect ourself from the social engineering attacks. **(9 MARKS)**

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